

IVAN SHABANOV

AI / Agent Engineer · AI Agents, Multi-Step LLM Pipelines & MCP · Evaluations, Human-in-the-Loop & Safe Tooling

Yerevan, Armenia (GMT+4) · Ready to relocate to Tallinn, Estonia · Remote until relocation, full EU overlap

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SUMMARY

AI Engineer with 4+ years of production engineering and 3 years shipping LLM systems and autonomous agents: built a confidence-based model cascade that cut API costs 10× and an AI conversational agent that tripled monthly channel revenue to ~\$141K after a 3-month controlled evaluation against human baselines. Engineered custom MCP servers exposing database schemas and task trackers, established project boundaries and safe tool constraints for Claude Code, and introduced an agentic pre-review stage that reduced module development time from ~2 h to ~40 min. Covers the full agent loop: prompt and routing design, cost and latency optimisation, regression suites, human-in-the-loop checkpoints, and auditability of event workflows across distributed microservices.

EXPERIENCE

Alfa-Dengi (Alfa-Bank ecosystem) via GIDFINANCE — Senior Full-Stack Engineer · Tech Lead

Sep 2024 – Present · 2 yrs · Remote

Consumer lending product in the ecosystem of one of Russia's largest private banks; delivered via outsourcing partner GIDFINANCE. Cross-functional product team; I ship e2e features from the backend to the interface.

- Introduced AI-agent workflows across the engineering team: built **custom MCP servers** exposing database schemas and internal APIs to give agents safe tooling; defined strict boundaries (guardrails preventing agents from modifying migrations or financial logic) and an **agentic pre-review stage**, reducing typical module delivery from **~2 h to ~40 min**.
- Launched a credit-report product that **turned profitable in its first week** (net profit **+40%**, EPC doubled from **~\$1.9 to ~\$3.8**) by leading a 3-engineer team through technical design, user flows, the API contract, the NBKI credit-bureau integration and a Node.js backend querying a **6M-row** scoring dataset.
- Eliminated notification loss** under peak load by moving notification delivery and domain-event processing to **Apache Kafka**; delivered live application-status updates to the client via **WebSocket** on top of those events.
- Enabled marketers to launch automated push campaigns **with zero engineering involvement** (**~\$80K/month** generated) by building a status-driven campaign builder on NestJS and React with frequency caps and daily per-user limits.
- Cut load time of a 100K-row lead-scoring dataset (9K+ new leads/day) from **8 s to under 1 s** by reworking the data-access path: database profiling, server-side pagination, PostgreSQL query and index optimisation, and caching.

Appbooster — Full-Stack Engineer (contract)

Nov 2023 – Aug 2024 · 10 mos · Remote

Mobile-app marketing platform: ~100 employees, 10,000+ apps promoted for clients in 26 countries.

- Made **100% of payouts auditable** and screened out duplicate and faked proofs by modelling the full task lifecycle from pick-up to payout behind a moderation queue with anti-fraud rules, processing **hundreds of completions a day**.
- Grew the task-and-rewards product to **1,000 daily active users in 3 months** by owning the full cycle: React user dashboard for paid tasks and proof submission, chatbot re-engagement notifications and a **live verification queue** for admins.

SODA — Full-Stack Engineer

Aug 2022 – Oct 2023 · 1 yr 3 mos · Remote

Lead-generation agency for residential developers and law firms (~60 employees). Owned the client-server side: APIs, database, interfaces.

- Tripled monthly channel revenue from ~\$47K to ~\$141K** and cut lead response time from ~8 min to **under 30 s** by leading an AI lead-processing assistant and running a 3-month controlled comparison (humans / humans + AI / AI only) that moved the channel fully to AI.
- Cut cost per processed lead 10×** by designing an LLM model cascade: a cheap model handles routine leads and escalates to a stronger model only on low confidence, protecting margins without degrading response quality.
- Migrated a legacy PHP monolith to **Node.js/TypeScript microservices** (Express, PostgreSQL): service decomposition, business-logic and API migration.
- Removed manual lead hand-off by building an integration hub syncing the agency CRM with partner developers' CRM systems across disparate external APIs.

OPEN-SOURCE PROJECT

vk-ai-bot-platform (MIT) — multi-tenant AI chatbot platform: OpenRouter LLM API, per-community prompt versioning, token-aware context windows, multi-step re-engagement workflows on BullMQ, event deduplication, admin analytics. TypeScript, Fastify, Drizzle ORM, PostgreSQL, Redis, React, Docker Compose, GitHub Actions.

SKILLS

AI Agents & LLM Systems: LLM integration (Anthropic API, OpenAI API, OpenRouter), AI agents, agentic workflows, MCP (Model Context Protocol) custom servers, prompt engineering, model cascading and routing, cost and latency optimisation, LLM evaluation against human baselines, human-in-the-loop validation, AI chatbots, Claude Code, AI code review

Languages: TypeScript, JavaScript (ES6+), Node.js, SQL

Backend & Architecture: NestJS, Express, Fastify, REST APIs, GraphQL, WebSocket, Prisma ORM, microservices architecture, event-driven systems (Apache Kafka, BullMQ), domain-driven design, system design

Data & Infrastructure: PostgreSQL, Redis, Apache Kafka, BullMQ, Docker, Kubernetes, AWS (EC2, S3, networking), GitLab CI/CD, GitHub Actions, OpenTelemetry, Grafana

Testing & Engineering Process: Playwright (E2E), Vitest, Jest, Supertest, code review, partner API integrations, Agile/Scrum, technical documentation, mentoring

EDUCATION & LANGUAGES

Southern Federal University — Applied Informatics, 2022–2024

English — professional working proficiency (daily in a distributed team) · Russian — native